

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions:

Tick (✓) the correct answer:

1.



Name the feature of Java depicted in the above picture.

- (a) Encapsulation (b) Inheritance
 - ☒ (c) Data Abstraction (d) Polymorphism
2. Which of the following can flow freely in Procedure Oriented Programming languages?
- (a) function ☒ (b) data (c) statement (d) instruction
3. Which of the following is a Procedure Oriented Programming language?
- (a) Java (b) C++ ☒ (c) Pascal (d) Eiffel
4. Procedure Oriented Programming follows:
- ☒ (a) Top-down approach (b) Down-top approach
 - (c) Random approach (d) Normal approach
5. Which is not an Object Oriented Programming language?
- (a) Python (b) Eiffel (c) Small talk ☒ (d) COBOL
6. Which OOP principle enables operating a TV set using a remote control without knowing its internal mechanism?
- ☒ (a) Data Abstraction (b) Encapsulation
 - (c) Insulation (d) Inheritance
7. Which of the following principles uses base and derived classes?
- (a) Encapsulation ☒ (b) Inheritance
 - (c) Polymorphism (d) Data hiding
8. What is the literal meaning of Polymorphism?
- (a) Using different operations for same purpose
 - ☒ (b) Using same operation for different purposes
 - (c) Using same data in different functions
 - (d) None of them
9. Which of the following is not true for Encapsulation?
- ☒ (a) Data and functions are loosely attached.
 - ☒ (b) Data and functions are wrapped together.
 - (c) Data is secured from getting mishandled.
 - (d) None of them.

10. Read the following text and choose the correct answer:

Machine language is the form of computer language in which instructions are coded in terms of binary digits (bits), i.e., in the form of zeroes and ones (0's and 1's). The computer understands the machine code directly to be operated by the CPU.

Which is the most suitable statement for Machine language from the following?

- (a) It is one of the types of high level language.
- (b) It is directly understood by the users.
- ✓(c) It is a machine dependent language.
- (d) It is easier to write the code in machine language.

II. Fill in the blanks:

1. An act of using essential features without including the background details is called data abstraction.
2. In an Object Oriented Programming, the importance is given to data items.
3. Wrapping of data and functions together as a single unit is called Encapsulation.
4. An has a unique identity which contains some characteristics and behaviour.
5. The objects may communicate with each other through
6. In POP, the global data are loosely attached to the
7. The process by which a class acquires the properties from another class is known as
8. In Object Oriented Programming, using a function for many purposes is termed as

III. State whether the given statements are TRUE or FALSE:

1. In assembly language, the instructions are given in terms of 0s and 1s.
2. PASCAL is a structure oriented programming language.
3. Machine language requires a translator to convert input instructions to be understood by the machine.
4. Compilers and Interpreters are referred to as language processors.
5. A process, according to which, a class acquires the characteristics from another class is called Encapsulation.
6. Procedure oriented program stresses on data.
7. C++ is also an object oriented programming language.
8. An object is identified by its characteristics.
9. Function is a set of objects that share the common state and behaviour.
10. Encapsulation keeps data safe from outside interference.

IV. Assertion and Reasoning:

Assertion (A): BASIC, Java, Python, etc. are some popular high level languages.

Reason (R): The instructions written in high level languages need language translators to be understood by the system.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

V. Application based questions:

Object Oriented Programming is an approach to standardise the program by creating partitioned memory area for both data and functions. It does not allow data to flow freely from one function to other. In this system, the complete problem is decomposed into a number of entities called objects. Some of the object oriented principles are given in the box as below:

Polymorphism

Encapsulation

Inheritance

Abstraction

Pick an appropriate principle for each of the descriptions given below:

- (a) It allows deriving a new class from existing class. The new class will have combined features of both the classes.
- (b) It promotes function overloading which is using a number of functions/ methods with the same name but with different parameters.
- (c) It allows grouping or merging data and functions as a unit such that the data of a particular object is used in the associated functions.
- (d) It is using some external components without knowing the internal mechanism.

VI. Name the following:

1. Two categories of Low Level languages

- (a)
- (b)

2. Two Procedure Oriented languages

- (a)
- (b)

3. Two structure oriented programming languages

- (a)
- (b)

4. Two examples of Data Abstraction in your daily life

- (a)
- (b)

5. Two examples of real world objects

- (a)
- (b)

6. Two basic principles of Object Oriented Programming

- (a)
- (b)

PART B – SUBJECTIVE QUESTIONS

I. Write short notes with an example (if applicable):

1. Object Oriented Programming
2. Data Abstraction
3. Encapsulation
4. Machine Language
5. Polymorphism
6. Procedure Oriented Programming
7. Inheritance
8. Assembly Language

[ICSE 2010]

[ICSE 2016]

[ICSE 2017]

II. Distinguish between:

1. Object Oriented Programming and Procedure Oriented Programming
2. High Level language and Low Level language
3. Compiler and Interpreter

III. Answer the following questions:

1. Enlist the features of Object Oriented Programming.
2. Which OOP principle implements function overloading? Explain. [ICSE 2007]
3. Write three benefits/preferences of Object Oriented Programming.
4. Mention two limitations of:
 - (a) Procedure Oriented Programming
 - (b) Object Oriented Programming
5. How is Encapsulation useful in Object Oriented Programming? Explain.
6. What are the advantages of High Level languages?

EXERCISES



PART A – COMPETENCY FOCUSED QUESTIONS

I. Multiple Choice Questions:

Tick (✓) the correct answer:

1.



What does the above illustration depict in Java?

- (a) Name of a bird
 - (b) DOS based platform for Java programming
 - ☒ (c) Window based platform for Java programming
 - (d) A web browser to execute Java programs
2. Applet programming is a feature of which language:
- (a) C
 - (b) C++
 - ☒ (c) Java
 - (d) BASIC
3. Which software company produced Java?
- (a) Apple
 - ☒ (b) Sun Microsystem
 - (c) Samsung
 - (d) Microsoft
4. What is meant by case sensitive language?
- ☒ (a) Uppercase and lowercase letters are distinguished.
 - (b) Lowercase letters are converted into uppercase letters.
 - (c) Uppercase letters are converted into lowercase letters.
 - (d) Uppercase and lowercase letters are treated to be same.
5. What is binary equivalent of source code?
- (a) ASCII code
 - (b) Unicode
 - (c) Octal code
 - (d) Byte code
6. A Java package which contains Scanner class to input values to the computer is:
- (a) java.io
 - (b) java.util
 - (c) java.awt
 - (d) java.net
7. A Java program is written under:
- (a) Sub-function
 - (b) Nested function
 - (c) Main function
 - (d) Overloaded function
8.statement displays the result in Java programming.
- (a) System.print()
 - (b) Print()
 - (c) System.printout()
 - (d) System.out.println()
9. Which keyword includes a package in a Java program?
- (a) new
 - (b) public
 - (c) import
 - (d) class
10. Read the following text and choose the correct answer:
BlueJ version is a menu driven approach for Java programming. It allows the users to carry out different tasks by clicking the options available on the menu bar.

Which is the most suitable statement for BlueJ from the following?

- (a) It is a window based platform.
- (b) It is a DOS based platform.
- (c) It is a window based and DOS based platform.
- (d) Every task is performed by typing the commands on the screen.

II. Write True or False:

- 1. Java application is a Java program which is developed by users.
- 2. James Gosling developed Java programming language.
- 3. Machine codes are expressed using alphanumeric characters.
- 4. Byte code is the program in binary form.
- 5. JVM stands for Java Visual Management.

III. Fill in the blanks with appropriate words:

- 1. Java is a sensitive language.
- 2. In Java, the package used to find the smallest of two numbers is
- 3. The words which are preserved with the system are called words and they can not be used as variable names in Java programming.
- 4. A single line comment is represented by the symbol in Java programming.
- 5. BlueJ is a based platform to operate Java programs.

IV. Assertion and Reasoning based questions:

Assertion (A): In Java, when you declare a class, it displays a sample program for the user.

Reason (R): You cannot write a Java program without the use of a sample program.

Based on the above discussion, choose an appropriate statement from the options given below:

- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is a correct explanation of Assertion (A).
- (b) Both Assertion (A) and Reason (R) are true and Reason (R) is not a correct explanation of Assertion(A).
- (c) Assertion (A) is true and Reason (R) is false.
- (d) Assertion (A) is false and Reason (R) is true.

V. Application based questions:

Java is such a programming language which allows creating logic using simple statements. It is also focused on web designing and string manipulation system even without calling default package. Based on the above discussion, following are some of the features or terms described. Now, answer these questions:

- (a) It is a Java program that is executed using Java based web browser.
- (b) It is a Windows based platform for Java programming.
- (c) It is a default package of Java programming.
- (d) It is the executable code suitable to a specific platform, converted from Java source code.

PART B – SUBJECTIVE QUESTIONS

I. Answer the following questions:

1. Who developed Java? What was it initially called?
2. Give a brief on the historical development of Java.
3. Mention at least four features of Java.
4. Define Byte code.
5. What is Java Virtual Machine (JVM)?
6. Name three packages of Java Class Library.
7. What are Java reserved words? Name any five.
8. Distinguish between JDK 1.3 and BlueJ.
9. A compiler is specific to a language. Give your comments.
10. What is the basic format of a Java Program? Explain with an example.
11. What is BlueJ?
12. Mention five features of BlueJ.
13. Name a package that is invoked by default. [ICSE-2007]
14. What are the points to be taken care of while naming a class in a Java program?
15. Java is a case sensitive language. Explain.
16. The main function in a Java program is declared as:

```
public static void main(String args[ ])
```


What is the significance of the words public, static and void? [ICSE-2007]
17. What does the term 'Compilation' mean?
18. A Java program uses a compiler as well as an interpreter. Explain.
19. Design a program in Java to display the following information on the output screen:
Name:
Class:
Roll No.:
Subject:
School:
20. You want to display your biodata on the output screen. Write a program in Java to perform the task in the given format:
Name:
Father's Name:
Date of birth:
Blood Group:
Aadhaar Card No.:
State: